

“CMB: Properties, Detection, and Research”

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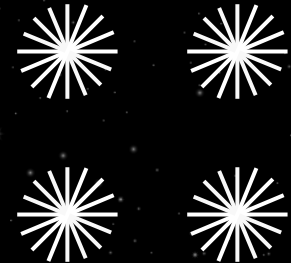
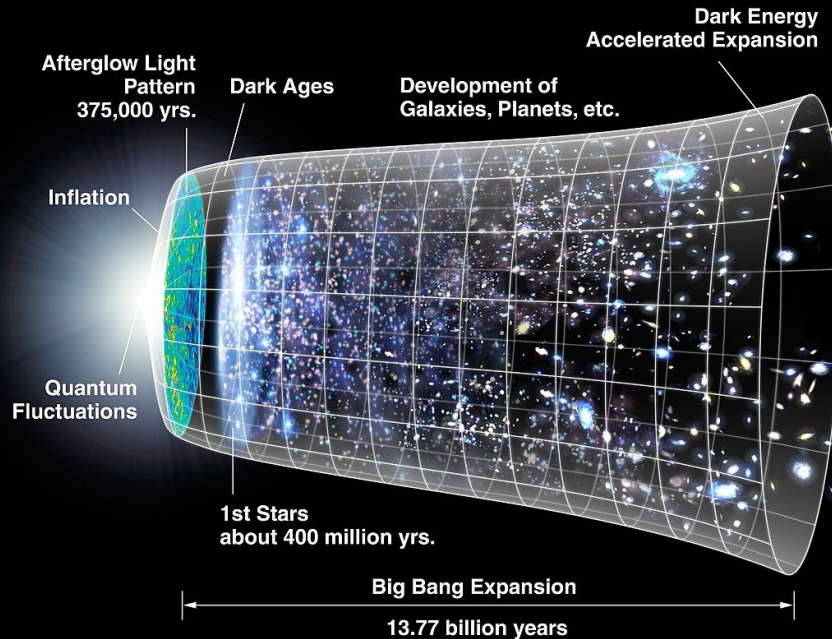


Observations and Models

01

Background

Cosmic History and Cosmological Parameters



[1] LAMBDA. NASA.

Cosmic History Background

- Big bang: extremely high density and high temperature state
- Ionized and opaque universe
- After $\sim 400,000$ years: epoch of recombination
- Radiation and matter decoupled
- 3,000K blackbody redshifted to observable faint background of microwaves

Λ CDM Parameters

- Λ CDM: mathematical model of Big Bang Cosmology
- Standard Λ CDM requires 6 independent variables to define the model

Independent parameters:

- Physical baryon density $\Omega_b h^2$
- Physical CDM density $\Omega_c h^2$
- Scalar Spectral Index n_s
- Optical depth to reionization τ
- Primordial curvature perturbations A_s
- Angular size of sound horizon θ_{MC}

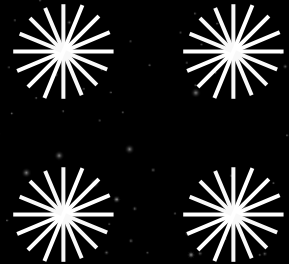
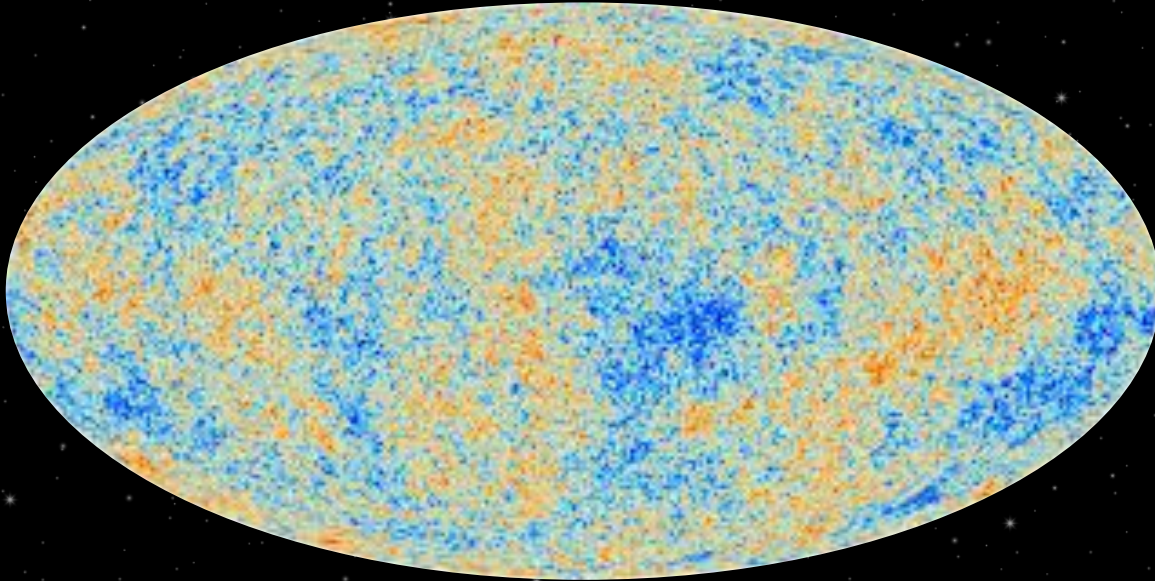
Derived values:

- Hubble constant H_0
- Dark Energy Density Ω_Λ
- Fluctuation amplitude σ_8
- Age of the Universe t_0

02

CMB Properties

Anisotropies, Power Spectrum, and Polarization



[2] *Planck*. COSMOS.

CMB Properties

- CMB: 2.725K Blackbody
- Overall isotropic
- Small anisotropies reflect early density perturbations
- Density perturbations → observable CMB temp fluctuations
 - Photon density fluctuations
 - Gravitational redshift of photons
 - Doppler effect from electrons scattering CMB photons

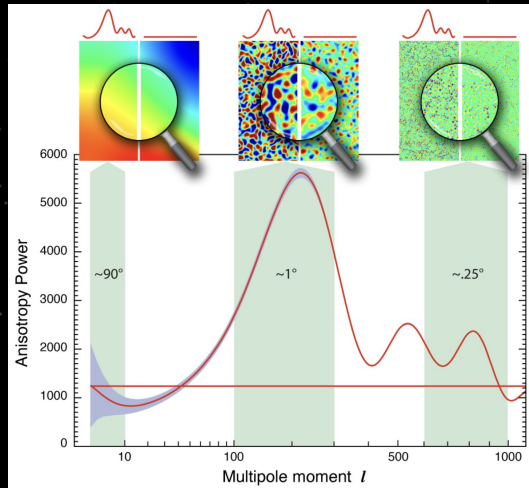
$$\frac{\Delta T}{T}(\vec{n}) \approx \frac{1}{4}\delta_{\gamma r} + \frac{1}{3}\frac{\Delta\phi_r}{c^2} - \vec{n} \cdot \frac{\vec{v}_r}{c}$$

[3] Bernardis and Masi (2012)

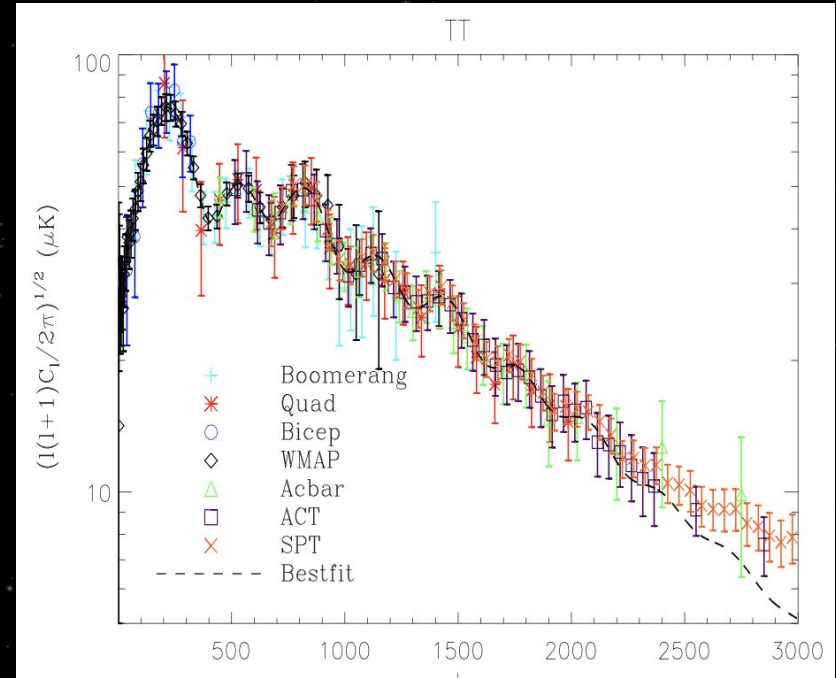


CMB Properties (cont.)

- Statistical description of fluctuations
- Angular power spectrum of CMB temperature anisotropy
- Multipoles correspond to different angular scales



[1] LAMBDA. NASA.

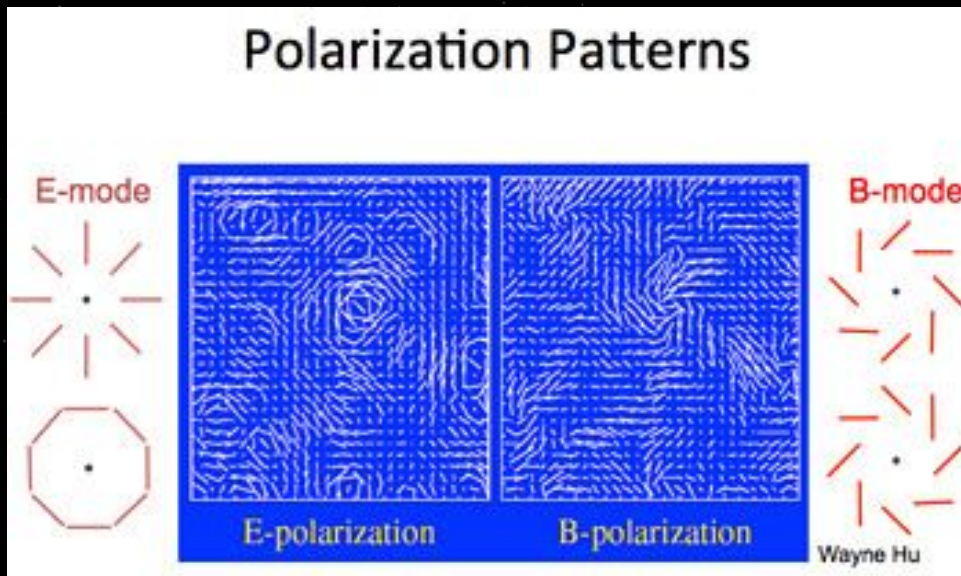
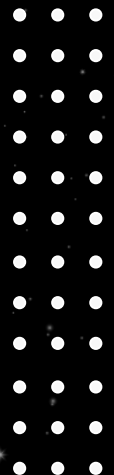


[3] Bernardis and Masi (2012)

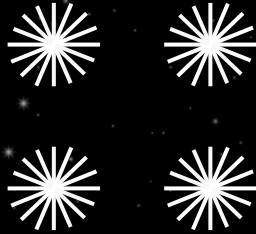


CMB Properties (cont.)

- CMB expected to be weakly polarized
- E-modes and B-modes
- Primordial gravitational waves leave imprint on CMB
- Presence of B-modes proves inflationary model



03

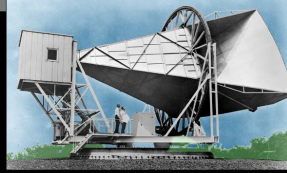


Detecting the CMB

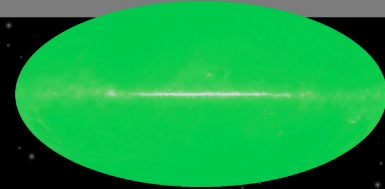
Methods, Telescopes, and Difficulties

[1] LAMBDA. NASA.

1965



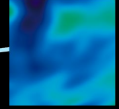
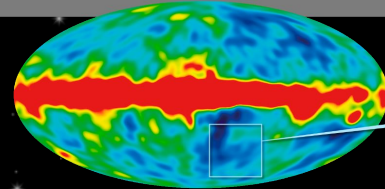
Penzias & Wilson



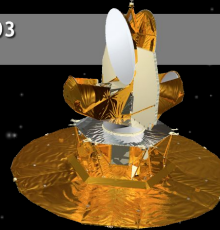
1992



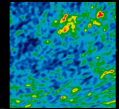
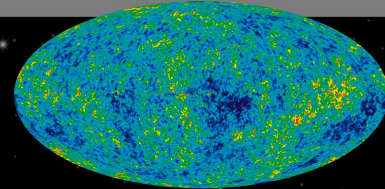
COBE



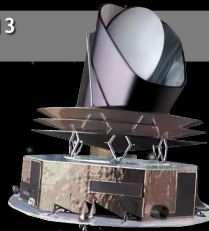
2003



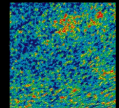
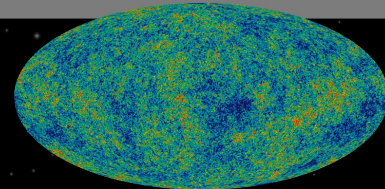
WMAP



2013



PLANCK



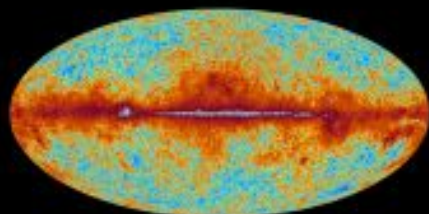
Detecting the CMB

- Greatest difficulty is reducing contamination
- CMB telescopes must have high angular resolution, sensitivity, and mapping speed
- Common features of telescopes measuring the CMB: Cryogenic cooling systems and Bolometric Detectors
- Ground-based (Simons Observatory), Stratospheric balloon (BLAST), or Satellite (Planck)
- CMB can be observed at different wavelengths or frequencies

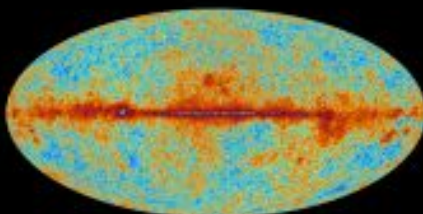


planck

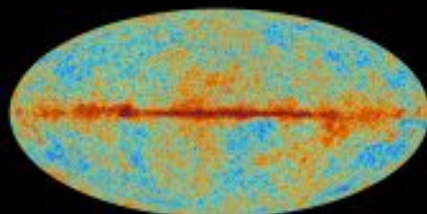
The sky as seen by Planck



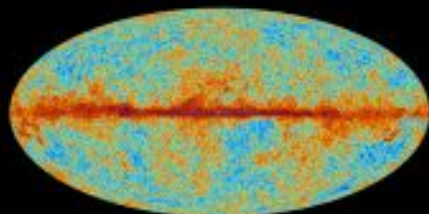
30 GHz



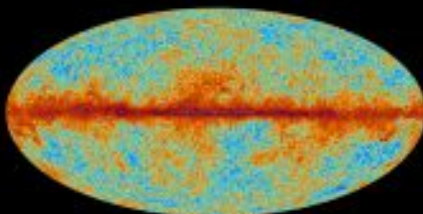
44 GHz



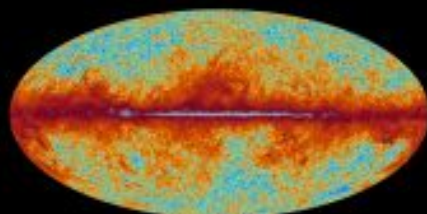
70 GHz



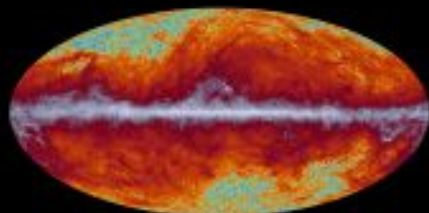
100 GHz



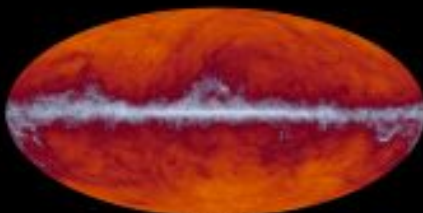
143 GHz



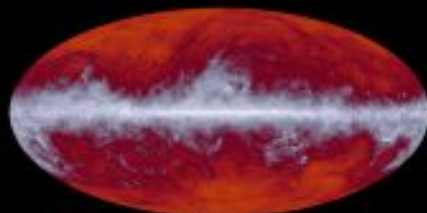
217 GHz



353 GHz

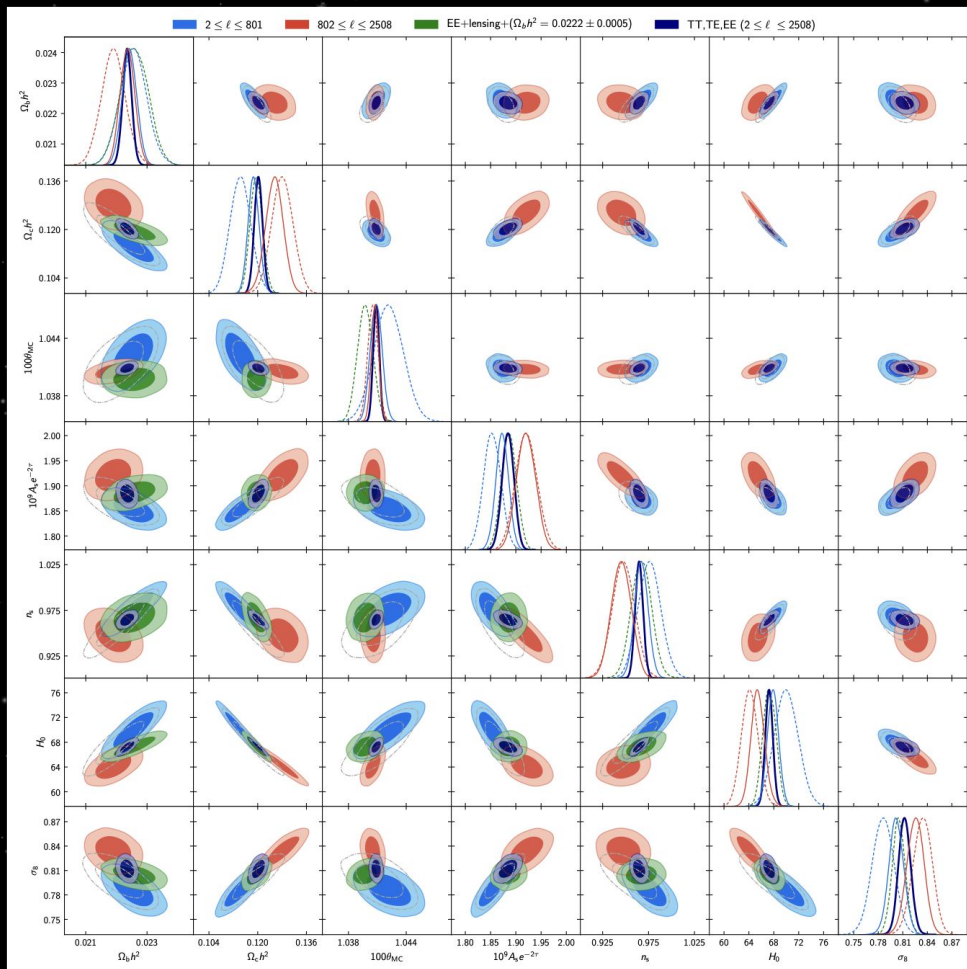


545 GHz

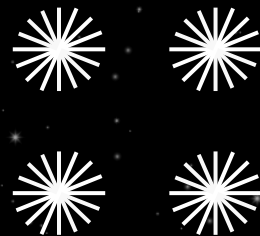


857 GHz

[2] Planck. COSMOS.



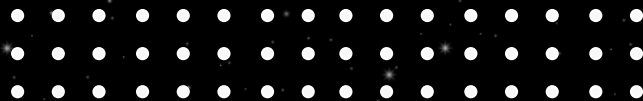
04



CMB Research

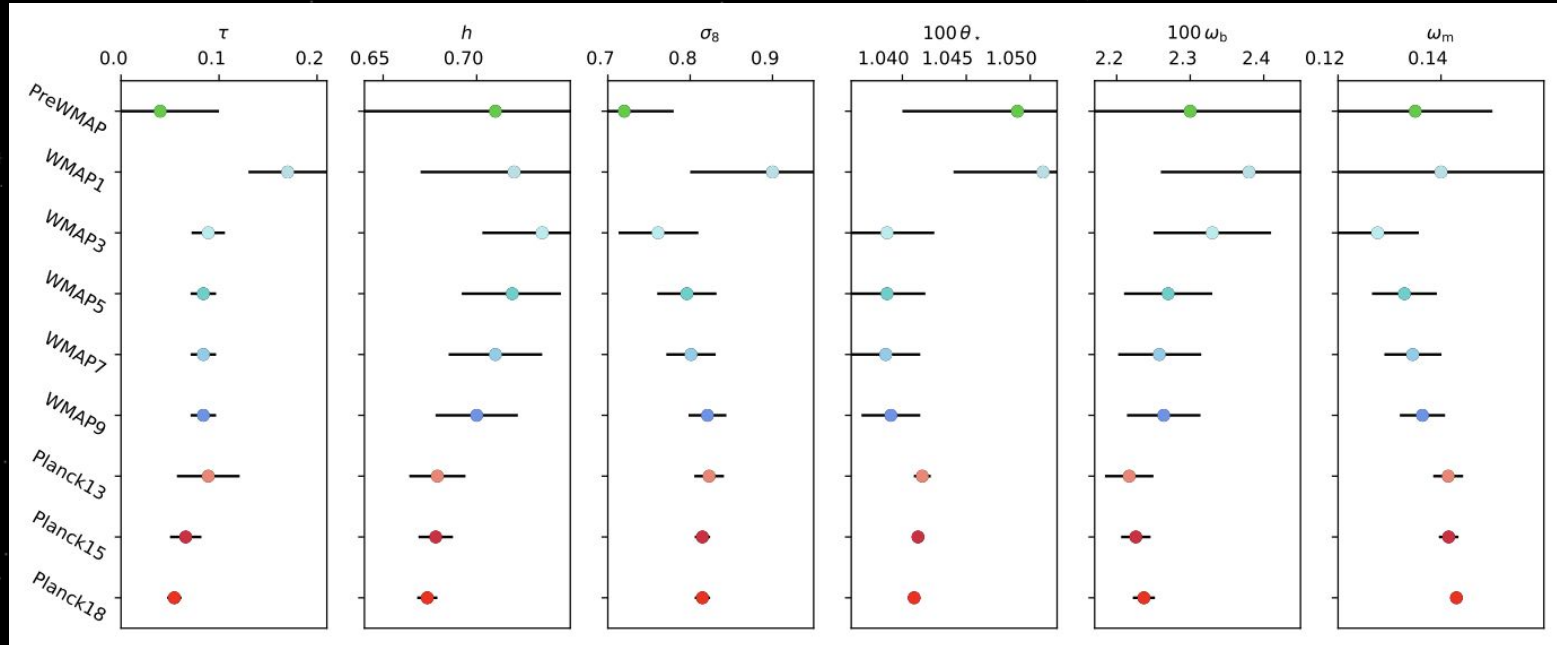
Observations and Models

[2] *Planck*. COSMOS.



CMB Research

- Constraining Parameters

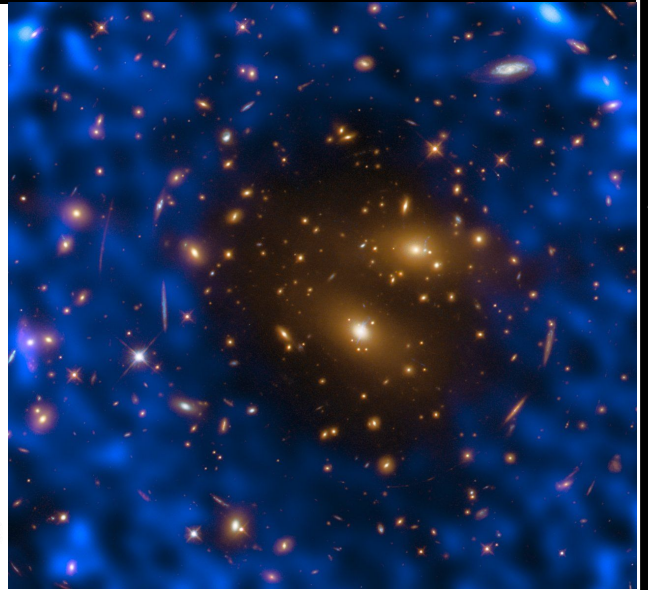
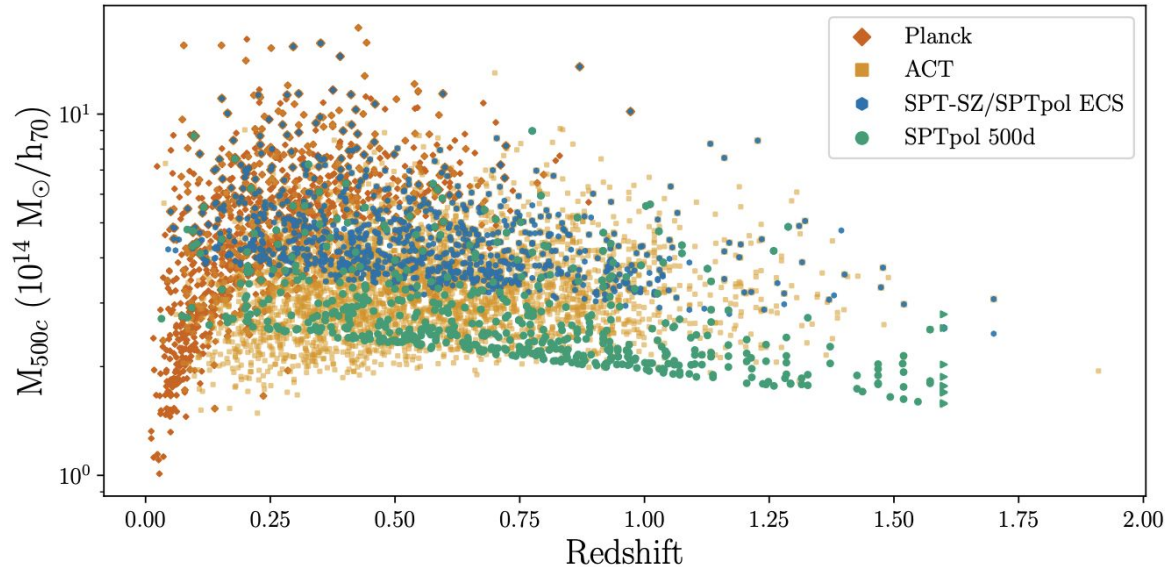


CMB Research (cont.)

- Anisotropy measurements of clusters of galaxies
- Sunyaev–Zel’dovich (SZ) effect: caused by high energy electrons scattering CMB photons
- Discover new clusters and study cluster physics

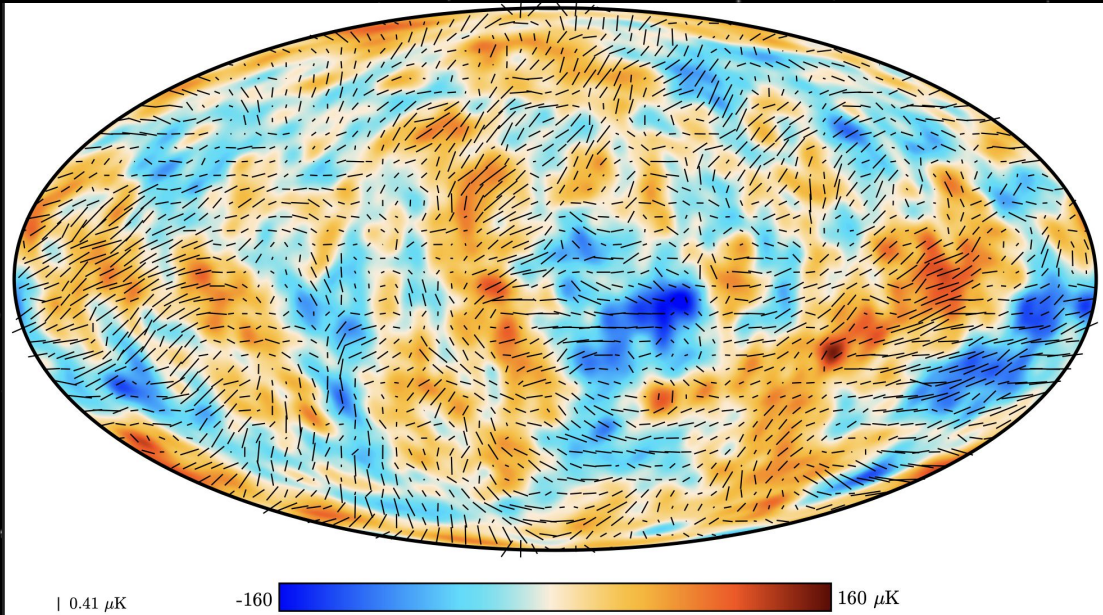
[4] Bleem et al. (2024)

[5] ALMA (2017)



CMB Research (cont.)

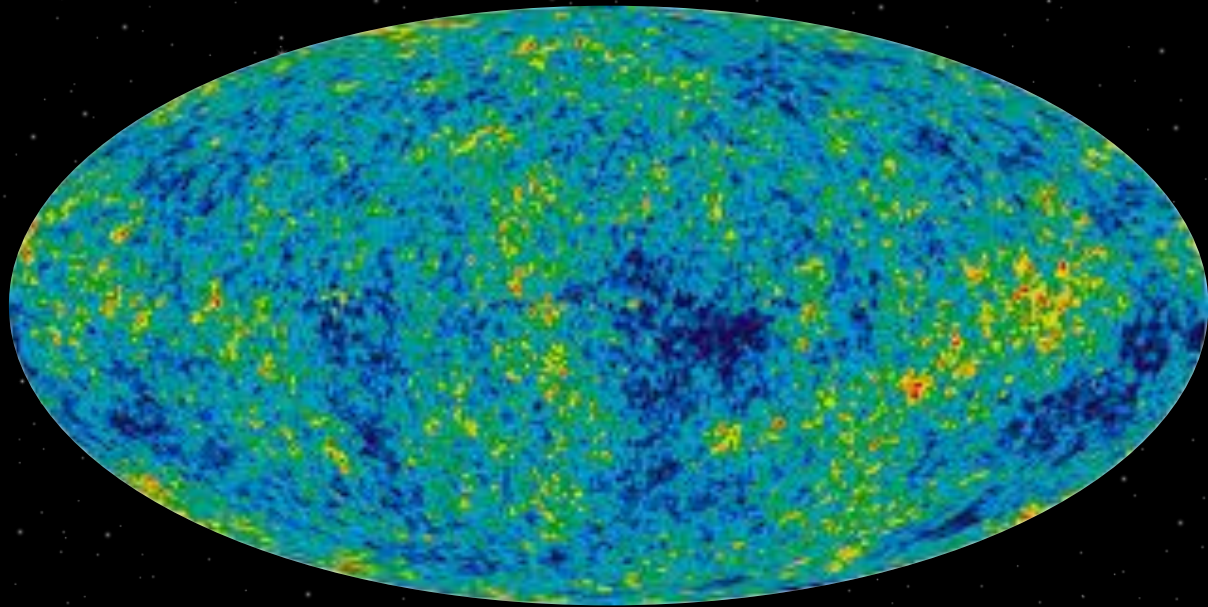
- Polarization Measurements
- Detection of B-modes



Conclusion

- CMB is a relic from the early universe
- Measurements of the CMB tells us information about the early universe and how the universe today is structured
- Studying the CMB requires precise measurements and advanced telescopes due to contamination
- CMB has been detected by several different telescopes and in different ways
- Current CMB research focuses on constraining parameters, studying galaxy clusters, and polarization measurements

Questions?



References

- [1] NASA. *Goddard Space Center*. Lambda - Legacy Archive for Microwave Background Data Analysis. <https://lambda.gsfc.nasa.gov/>
- [2] ESA. *COSMOS*. Planck Science Team Home. <https://www.cosmos.esa.int/web/planck>
- [3] Bernardis, P., & Masi, S., (2012). The cosmic microwave background: observing directly the early universe. *Space Telescopes and Instrumentation 2012: Optical, Infrared, and Millimeter Wave*, 8442 (0277-786X), 844203. <https://arxiv.org/abs/1208.0298v1>.
- [4] Bleem et al., (2024). Galaxy Clusters Discovered via the Thermal Sunyaev-Zel'dovich Effect in the 500-square-degree SPTpol Survey. *The Open Journal of Astrophysics*, 7 (2565-6120). <https://arxiv.org/abs/2311.07512>.
- [5] ALMA. (2017, March 17). *ALMA's ability to see a "Cosmic Hole" confirmed*. ALMA Telescope. https://alma-telescope.jp/en/news/mt-almas_ability_to_see_a_cosmic_hole_confirmed.html.

